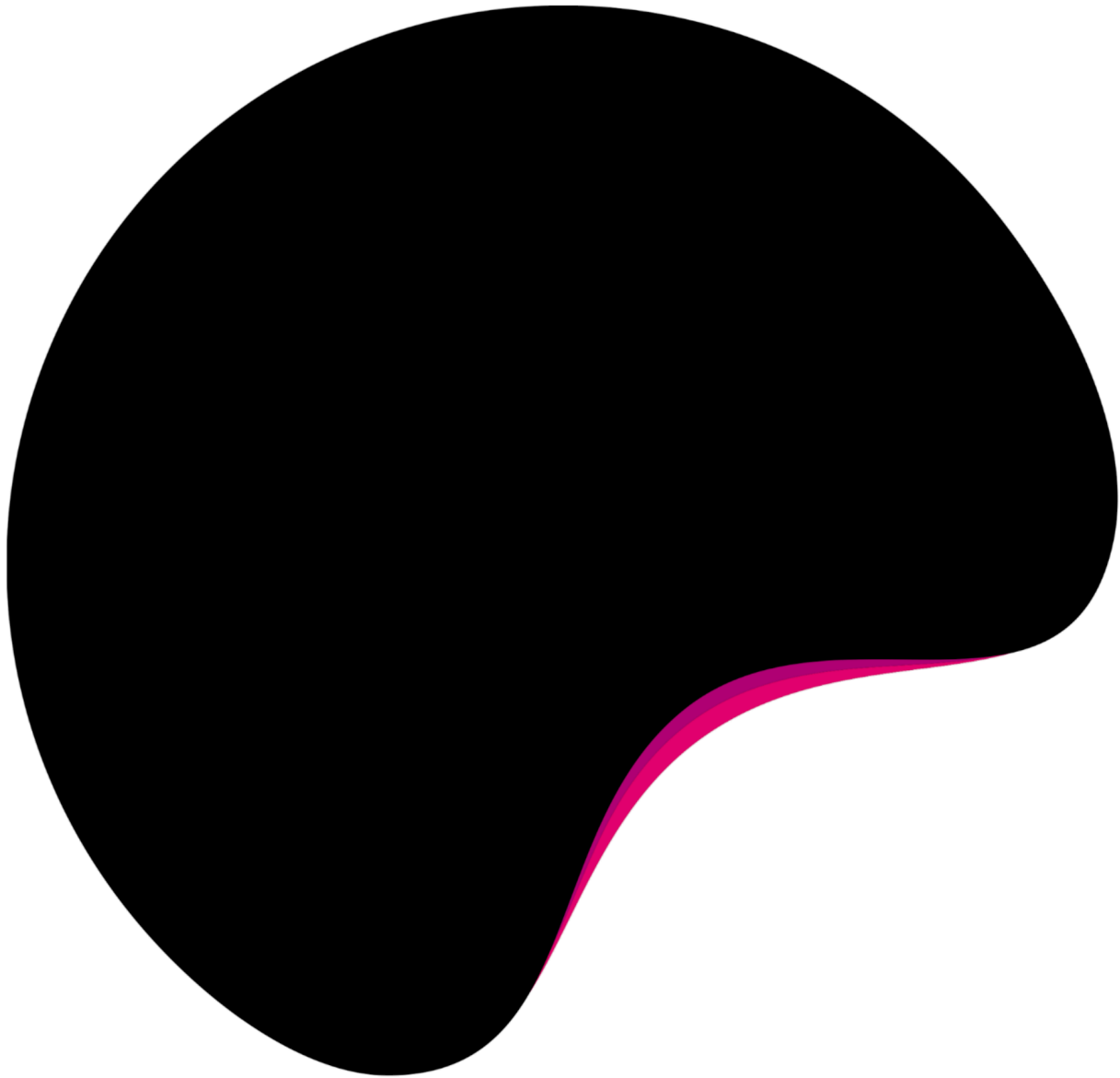




GlowCore

SE Project Documentation

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Part I

Management Summary

Management Summary

Part II

Product Documentation

1 Glossary

Term	Definition
Grid	Refers to the 2D Grid that divides the GlowCore world into rows and columns.
Tile	A Tile is a single unit of space in the grid and has the form of a square.
Node	A Node refers to an object that is occupying a Tile. Not every Tile has a Node, but every Node belongs to a Tile.
World Item	World Items are items that can be picked up by the player and physically exist somewhere in the GlowCore world. In contrast to Nodes they do not belong to a Tile and do not have a hitbox.
Inventory Item	Inventory Items refer to the items that the player possesses. Certain Nodes such as Chests can also hold Inventory Items

Table 1: Glossary

2 Requirements

2.1 Functional Requirements

2.1.1 Actors / Personas

Since a game typically only has one primary actor - **The Player** - it makes sense to define Personas that reflect different types of players. The categorization into Casual- and Dedicated Players covers most player types. In the following, the pronoun **he** is used for simplicity, but the Personas are meant to include all players.

2.1.1.1 Casual Player

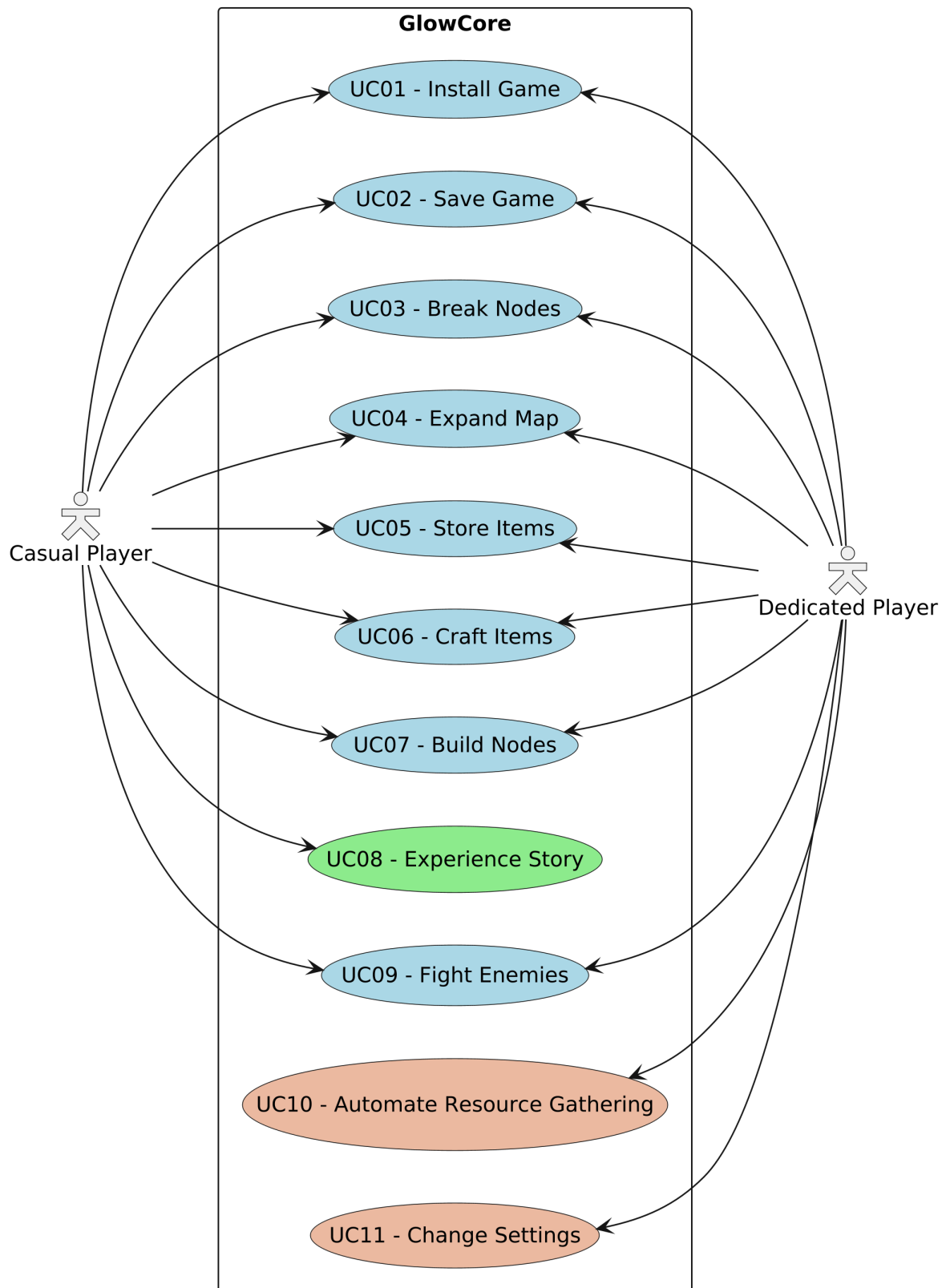
- The Casual Player plays GlowCore infrequently and for short amounts of time. He wants to relax and unwind from his life. He prefers to play the game just for fun without putting effort into it.
- The Casual Player wants to experience the game from beginning to end with a story and explore the world. When he plays the game for the first time, he depends on the game to explain the rules of the game to him without being too overwhelming and tedious. As soon as he finishes the game, he will be done with it, having experienced everything he wanted.
- The Casual Player never changes settings in games, he never feels the need to customize anything. He likes to play the game as it is.

2.1.1.2 Dedicated Player

- The Dedicated Player plays GlowCore in long sessions. He loves games where you build factories and automate everything. He searches for the feeling of progression and finds satisfaction in it. He always searches for a better way to progress and optimize his gameplay.
- The Dedicated Player has already played the game multiple times before, thus he doesn't want to be bored with a long tutorial everytime he starts a new game. He wants to be able to dictate the pacing of the game.
- He doesn't care for the story and lore in games. He doesn't want to read through dialogues and doesn't like long cutscenes.

2.1.2 Use Case Diagram

The diagram shows a modified version of the Use Case Diagram. The Personas take the place of the Actors. An arrow indicates that a Persona uses a feature. A missing arrow means that the Persona is not interested in using a feature, although it remains accessible to him.

**Image 1:** Use Case Diagram

2.1.3 Use Cases

The Use Cases are described in the brief format, with each Use Case including an appropriately detailed description.

It is essential for the project that all Use Cases that define the **MVP** are implemented. The Use Cases marked with **Priority: Middle** are planned to be implemented, but can be reduced if the MVP takes more time than anticipated. The Use Cases marked with **Priority: Low** will be implemented if enough time is left.

ID	UC01
Name	Install Game
Persona	Casual Player & Dedicated Player
Priority	High - Required for MVP
Description	The player can install the game. The installation requires no technical skills.

Tab. 2: UC01 - Install Game

ID	UC02
Name	Save Game
Persona	Casual Player & Dedicated Player
Priority	High - Required for MVP
Description	When the player stops playing, the game saves his save state in a file and loads it when he resumes the game.

Tab. 3: UC02 - Save Game

ID	UC03
Name	Break Nodes
Persona	Casual Player & Dedicated Player
Priority	High - Required for MVP
Description	The player can break Nodes in the world with his hands or with tools. The usage of tools e.g. an axe to break a tree, increases the speed significantly. When the Nodes break, they drop items depending on the type of Node. E.g. a Tree drops a Wood-Item. Certain Nodes that are built by the player themselves such as a workbench drop themselves upon breaking.

Tab. 4: UC03 - Break Nodes

ID	UC04
Name	Expand Map
Persona	Casual Player & Dedicated Player
Priority	High - Required for MVP
Description	The player is confined to only move on tiles that are unlocked at a given point. The player can drop off items at the GlowCore, whereupon the map expands. The player can then move into and explore the new area.

Tab. 5: UC04 - Expand Map

ID	UC05
Name	Store Items
Persona	Casual Player & Dedicated Player
Priority	High - Required for MVP
Description	Items that the player picks up are stored in his inventory. The player's inventory has a limited size. To organize items, the player can move items into chests and take them out again.

Tab. 6: UC05 - Store Items

ID	UC06
Name	Craft Items
Persona	Casual Player & Dedicated Player
Priority	High - Required for MVP
Description	The player can craft items into other items with predefined recipes. Without a workbench the player is limited to a few simple recipes. The player can interact with a workstation such as a workbench to unlock more crafting recipes. Workstations such as an oven require additional items: e.g. Coal to smelt an Ingot.

Tab. 7: UC06 - Craft Items

ID	UC07
Name	Build Nodes
Persona	Casual Player & Dedicated Player
Priority	High - Required for MVP
Description	The player can place certain items onto the world, where they become Nodes. Each Tile can hold only one Node, and the player is not allowed to build a Node on an already occupied Tile.

Tab. 8: UC07 - Build Nodes

ID	UC08
Name	Experience Story
Persona	Casual Player
Priority	Low
Description	The player can experience a story while playing the game. When the game starts there is an intro sequence. There is an end to the game, at which point a cutscene is played. The game contains small narrative hints that fit into the story conveyed by the GlowCore-World, including the acoustic, the visual art style and the overall gameplay experience.

Tab. 9: UC08 - Experience Story

ID	UC09
Name	Fight Enemies
Persona	Casual Player & Dedicated Player
Priority	Middle
Description	The player can engage in combat with enemies. Defeated enemies drop items that are useful for the player. When the enemy hits the player, he loses hitpoints. When the player has no hitpoints left, he loses some of his items and respawns near the center of the map.

Tab. 10: UC09 - Fight Enemies

ID	UC10
Name	Automate Resource Gathering
Persona	Dedicated Player
Priority	Middle
Description	The player can automate tedious tasks, such as breaking Nodes. He can craft machines that are placed as Nodes and they automatically perform a task e.g. chopping down trees. There is a selection of machines that can be combined to automate more and more. Example: The player places multiple Replanter-Machines in a row which automatically replant trees on the neighbouring tile. The player places an Axe-Machine onto the start of the line, at which point the Axe-Machine chops one tree after another and then moves onto the next tile. To automate it even further, the player places Rotaters on both which cause the Axe-Machine to turn around on contact.

Tab. 11: UC10 - Automate Resource Gathering

ID	UC11
Name	Change Settings
Persona	Dedicated Player
Priority	Middle
Description	The player can change settings of the game, such as the FOV and the Volume.

Tab. 12: UC11 - Change Settings

2.2 Non-Functional Requirements

3 Domain Analysis

4 Architecture

5 Quality Measures

5.1 Working Environment

5.2 Testing

Part III

Project Documentation

6 Initial Project Proposal

Project name: GlowCore

Date: 19.12.2025

6.1 Team Members

1. Cedric Cathomas (Cedric.Cathomas@ost.ch)
2. Dominik Wyss (Dominik.Wyss1@ost.ch)
3. Nathanael Fässler (Nathanael.Faessler@ost.ch)
4. Yoris Kucera (Yoris.Kucera@ost.ch)

6.2 Availabilities

Time slot	Mon	Tue	Wed	Thu	Fri
08h00–09h00	-	(XR)	-	-	(XO)
09h00–10h00	-	(XR)	-	-	(XO)
10h00–11h00	(XR)	-	-	-	(XO)
11h00–12h00	(XR)	-	-	-	(XO)
12h00–13h00	-	-	-	-	(XO)
13h00–14h00	XR	-	-	-	(XO)
14h00–15h00	XR	-	-	-	(XO)
15h00–16h00	(XR)	XR	-	-	(XO)
16h00–17h00	(XR)	XR	-	-	(XO)
17h00–18h00	-	(XR)	-	-	(XO)
18h00–19h00	-	(XR)	-	-	-

Table 13: Team availability for review meetings

6.3 Vision

GlowCore is a singleplayer progression game focused on resource gathering and gradual world expansion. At the start, the player can only access a small, limited area of the game world, while the rest remains hidden and unreachable. At the center of the world is a special entity that can be upgraded using collected materials. Each upgrade expands the accessible area, revealing new parts of the world and enabling further progression. The core gameplay loop is built around exploring newly unlocked areas, gathering resources, and using them to continuously improve the central entity and push the boundaries of the world outward.

As the world expands, additional gameplay systems deepen the experience. The player can obtain and use different materials, interact with tools, craft items, and face enemies within the expanding world. More advanced mechanics such as machines for processing resources, farming systems, and specialized equipment can be introduced over time to increase complexity and strategic depth.

6.4 Proposed Realisation

The project will be implemented using Unity as the primary game engine, using its integrated systems for rendering, physics, input handling, animation, audio, and crossplatform deployment. Development will be done in C# using Unity's component-based architecture (GameObjects, MonoBehaviours, and ScriptableObjects) to structure gameplay logic and systems. Unity's built in physics engine and animation system (Animator) will be used to realise core gameplay and user interaction. Visual assets will be rendered using Unity's Universal Render Pipeline (URP). Individual systems will be tested for robustness using unit tests by method of Unity's own testing framework, based on NUnit. Version control will be handled with Git. The project will be developed within the Unity Editor.

7 Project Plan

7.1 Resources

7.1.1 People

The project team consists of 4 students. Each member contributes approximately 8.6 hours per week and is responsible for tasks of their defined roles.

7.1.1.1 Project Management Roles

Role	Description	Names
Product Owner	• Define product vision • Manage backlog • Prioritize features • Validate requirements	Nathanael Fässler
Scrum Master	• Facilitate meetings • Remove blockers • Ensure Scrum process • Support team workflow	Dominik Wyss
Project Manager	• Schedule planning • Resource management • Risk control • Progress reporting	Cedric Cathomas
DevOps Engineer	• CI/CD maintenance • Build automation • Deployment support • Infrastructure management	Yoris Kucera

Table 14: Project Management Roles

7.1.1.2 Developer Roles

Role	Description	Names
Lead Developer	• Overall technical coordination • Code quality control • Architecture decisions • Integration management	Yoris Kucera
Developer	• Gameplay implementation • Bug fixing • Feature development • System integration • Writing Test Cases	Dominik Wyss Yoris Kucera Nathanael Fässler Cedric Cathomas
Art Director / 3D Artist	• Asset creation • Scene design • Visual consistency • Asset optimization	Nathanael Fässler
UI Architect	• UI/UX design • Menu systems • HUD implementation • User interaction flow	Cedric Cathomas
Graphics Programmer	• Shader development • Particle effects • Performance optimization • Visual effects integration	Yoris Kucera
Audio Engineer	• Sound design • Audio integration • Music management • Audio balancing	Dominik Wyss
Level Designer	• Level layout • Gameplay balancing • Difficulty tuning • Playtesting feedback	Nathanael Fässler Cedric Cathomas

Table 15: Developer Roles

7.1.2 Time

The project runs from 16.02.2026 to 12.06.2026 with an estimated effort of 120 hours per person, resulting in a total workload of 480 hours, structured in milestones and sprints.

7.1.3 Cost

Most required tools and platform are freely available through free or student licenses. We plan on publishing GlowCore on steam for which steam charges a 100 US-Dollar publishing fee. The cost will be shared by the members.

7.2 Organization

7.2.1 Meetings

The project meetings follow mainly the Scrum standard. With weekly Scrum meetings replacing daily stand-ups and biweekly sprint planning including retrospectives. Also the Regular advisor meetings to present and discuss our results so far.

Type	Todos	Frequency
Sprint Review (Advisor)	• Present sprint results • Demonstrate features • Discuss open issues • Collect advisor feedback • Document decisions	Every 2 weeks Tuesday 15:00 - 16:00
Weekly Scrum	• Report progress • Define next steps • Identify blockers • Update tasks • Support team members	Weekly Friday 14:00 - 15:00
Sprint Retrospective	• Analyze sprint process • Identify problems • Define improvements • Document lessons learned	Every 2 weeks Friday 15:00 - 16:00
Sprint Planning	• Define sprint goals • Select backlog items • Estimate effort • Assign tasks • Update schedule	Every 2 weeks Friday 16:00 - 17:00 (after retro)

Table 16: Planned Meetings Overview

7.2.2 Planning Tools

Tool	Use Case
GitHub	• Version control • GitHub Actions (CI/CD) • Pull requests • GitHub notifications via Discord
Typst	• Project documentation • PDF generation
Microsoft Teams	• File sharing • Communication
Jira	• Task management • Sprint planning • Backlog tracking • Issue tracking • Time tracking
Miro	• Brainstorming • Diagrams • Project planning
Discord	• Online meetings • Informal communication • Quick coordination • GitHub activity notifications

Table 17: Planning Tools

7.3 Schedule

The project follows the OST internal SCRUM+ framework for its overall organization. SCRUM+ combines elements of Scrum with the Rational Unified Process (RUP). Long-term planning is based on the RUP approach, while short-term development is organized using Scrum with two-week iterations.

At the start of each iteration, we estimate all tickets with storypoints. The story **GC-54 Make Player** is the reference story which is worth 5 story points.

7.3.1 Work Items

We work with 5 types of Work Items

Name	Description
Epic	A group of features that we want to implement.
Story	Describes a feature that is directly related to the game
Task	Everything that isn't directly related to the game
Bug	Bugs are created when a bug is found and fixed in an unplanned matter.
Documentation	Work related to the documentation

Table 18: Work Items

7.3.2 Long Term Plan

The Long Term Planning is separated into two categories:

- Milestones: Serve as checkpoint for releases and review meetings.
- Epics: Provide the structure to organize the stories.

7.3.2.1 Milestones

The milestones for the review meetings and for the releases are interleaved and independent.

Nr	Milestone Name	Date	Description
M01	Review 1	10.03.2026	Initial Project Setup
M02	Review 2	24.03.2026	Requirements
M03	Review 3	14.04.2026	End of Elaboration
M04	Review 4	28.04.2026	Quality
M05	Review 5	12.05.2026	Architecture
M06	Project Presentation	12.06.2026	Final Presentation to SEProject-Advisors and fellow students. Completion of SEProject.
M07	Pre-Alpha Release	20.03.2026	First playable prototype is ready. Core gameplay loops are implemented. The player can chop trees, pick up wood and expand the GlowCore.
M08	Alpha Release	10.04.2026	MVP is done and working. Core gameplay Mechanics are implemented. Player can craft, build, expand the map and he has an inventory. The player can save his savestate.
M09	Beta Release	15.05.2026	More Biomes and Materials are added. There is a combat system with enemies and peaceful mobs. There is a title screen and a settings menu. Automation mechanics are implemented. There are more GlowCore Phases.
M10	Official Release and Final Submission	05.06.2026	The game is polished and has no mayor bugs. The game has a beginning and an ending.

Table 19: Milestones

7.3.2.2 Epics

Epic Name	Description/Requirements
Documentation	This epic includes all documentation related work.
Project Setup	This epic contains all work that is done in the Inception-Phase.
Core Gameplay Loop Mechanics	- The player can chop down trees, collect wood and feed it into the GlowCore. - The GlowCore levels up and expands the map. - The player is restricted to the unlocked map. The grid system is implemented.
Inventory, Crafting, Building	- Inventory: The player has an inventory and can store a limited amount of items. He can have items in his hands and in his backpack(or similar). - Crafting: The player can craft items to other items on different workstations. - Building: The player can place objects onto the world-grid.
Graphics	- The Artstyle of the game ist set. - The 3D Models follow a consistent Artstyle. - A Shader is implementet.
Game Framework & Completeness	This epic contains everything that makes a game complete and feel like a game. - Intro: When the Game starts the GlowCore Logo is displayed. After the title screen a Cutscene is played that leads into the game. - Ending: A ending of the game is defined. When the player finishes the game, a Cutscene is played and the credits roll. - UI: A Title Screen is available. In-Game there is a Debug Screen, and a Settings Menu. - Gamedata Saving: The player can save his progress.
Big World & GlowCore Phases	- World building: The game has a complete world to play in. - GlowCore Phases: The GlowCore has multiple phases, all of which need different materials to level up. The GlowCore changes visually with every Level.
Biomes & Materials	- The world has multitple biomes: e.g. Forest, Grassland, Mountain, Volcano, Jungle etc. - Each biome has unique ressources. - There are Caves which you can enter.
Combat	- Enemies can spawn and fight with the player. - The player can take damage and can die. - A respawn mechanic is set up. - There are peacuful mobs.
Resource Automation	- The Player can replant trees. - The Player can craft machines that automatically gather resources.
Playtesting & Release	- Playtesting: The game is testet by players that play it for the first time (blind test). - Release: The game is released on steam on its own steam page and has a trailer, screenshots and a description.

Table 20: Epics

The following image shows the Timeline in Jira as of 04.03.2026.

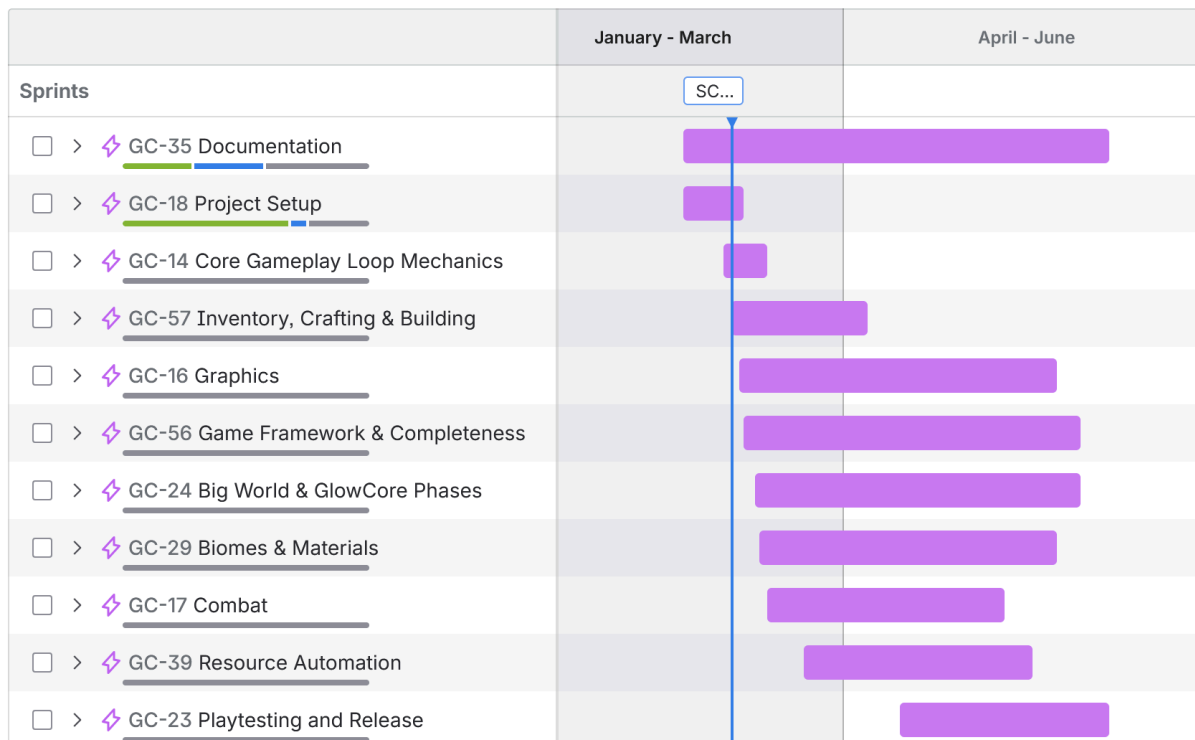


Image 2: Jira Timeline

7.3.2.3 Initial Backlog & MVP

The priority of Stories is defined by the ordering in the backlog. As we progress in Agile, we may divide Stories further and add new Stories. The MVP is reached when every Story up to and including **GC-101 Build World** is completed.

The following is the Initial Backlog as of 04.03.2026.

☐ Backlog (44 work items)				0 0 0	Create sprint
GC-54	There is a player	CORE GAMEPLAY LO...	TO DO	-	
GC-78	Create Tree-Node 3D Model	GRAPHICS	TO DO	-	
GC-95	Create Wood-Item 3D Model	GRAPHICS	TO DO	-	
GC-47	Trees can be chopped	CORE GAMEPLAY LO...	TO DO	-	
GC-109	Make GlowCore Phase 1 3D Model	GRAPHICS	TO DO	-	
GC-46	GlowCore expands World	CORE GAMEPLAY LO...	TO DO	-	
GC-69	Make Inventory-System	INVENTORY, CRAFTI...	TO DO	-	
GC-74	Save Game-Data	GAME FRAMEWORK &...	TO DO	-	
GC-108	Create Axe 3D Model	GRAPHICS	TO DO	-	
GC-97	Create Coal 3D Model	GRAPHICS	TO DO	-	
GC-106	Add Coal Node and Item	BIOMES & MATERIALS	TO DO	-	
GC-96	Create Stone 3D Model	GRAPHICS	TO DO	-	
GC-93	Add Stone Node and Item	BIOMES & MATERIALS	TO DO	-	
GC-107	Make Pickaxe 3D Model	GRAPHICS	TO DO	-	
GC-102	Add Pickaxe	BIOMES & MATERIALS	TO DO	-	
GC-79	Add Grassland Biome	BIOMES & MATERIALS	TO DO	-	
GC-111	Make GlowCore Phase 2 3D Model	GRAPHICS	TO DO	-	
GC-100	Add GlowCore Phase 2	BIG WORLD & GLOWC...	TO DO	-	
GC-71	Make Crafting-System	INVENTORY, CRAFTI...	TO DO	-	
GC-70	Make Building-System	INVENTORY, CRAFTI...	TO DO	-	
GC-77	Implement Pixelize-Shader	GRAPHICS	TO DO	-	
GC-103	Add Chest	INVENTORY, CRAFTI...	TO DO	-	
GC-101	Build World	BIG WORLD & GLOWC...	TO DO	-	
GC-99	Add Rock Biome	BIOMES & MATERIALS	TO DO	-	
GC-72	Game Intro	GAME FRAMEWORK &...	TO DO	-	
GC-73	Game Ending	GAME FRAMEWORK &...	TO DO	-	
GC-105	Add Combat Logic	COMBAT	TO DO	-	
GC-124	Create Enemy 3D Model	GRAPHICS	TO DO	-	
GC-81	Add Enemy	COMBAT	TO DO	-	
GC-75	Ingame Debug Settings	GAME FRAMEWORK &...	TO DO	-	
GC-76	Player Settings-Menu	GAME FRAMEWORK &...	TO DO	-	
GC-125	Create Peaceful-Mob 3D Model	GRAPHICS	TO DO	-	
GC-82	Add peaceful Mob	COMBAT	TO DO	-	
GC-123	Make Sword 3D Model	GRAPHICS	TO DO	-	
GC-83	Add Sword	COMBAT	TO DO	-	
GC-120	Make Axe-Machine 3D Model	GRAPHICS	TO DO	-	
GC-114	Add Axe-Machine	RESOURCE AUTOMAT...	TO DO	-	
GC-121	Make Replant-Machine 3D Model	GRAPHICS	TO DO	-	
GC-115	Add Replant-Machine	RESOURCE AUTOMAT...	TO DO	-	
GC-122	Make Rotater-Block 3D Model	GRAPHICS	TO DO	-	
GC-116	Add Rotater-Block	RESOURCE AUTOMAT...	TO DO	-	
GC-117	Conduct Playtesting with User	PLAYTESTING AND R...	TO DO	-	
GC-118	Release Game on Steam	PLAYTESTING AND R...	TO DO	-	
GC-119	Make Marketing Material for Steam Page	PLAYTESTING AND R...	TO DO	-	

Image 3: Inital Backlog

7.3.3 Short Term Plan

In a Sprint the priority of the tasks and stories is defined by the ordering.

7.3.3.1 Sprint 1 (Sprint Zero) 20.02.2026 - 06.03.2026

The following shows the Jira sprint backlog of the first sprint. Since this was Sprint Zero, not all guidelines were followed initially because they had not yet been established.

Work	Assignee	Status
 GG-88 Document Long and Short Term Plan	 Nathanael Fässler	DONE 
 GG-135 Document Sprint Planning Meeting Sprint 2	 Nathanael Fässler	DONE 
 GG-92 Setup Contract with OST about Ownership	 Nathanael Fässler	DONE 
 GG-89 Document Meeting Weekly Scrum Sprint 1	 Nathanael Fässler	DONE 
 GG-4 Concept for Code Quality, Linter etc.	 Yoris Kucera	DONE 
 GG-3 Unity Setup	 Yoris Kucera	DONE 
 GG-126 Documentation list of tables	 Cedric Cathomas	DONE 
 GG-1 Risk Specification	 Yoris Kucera	DONE 
 GG-91 Document Guidelines for code and documentation	 Yoris Kucera	DONE 
 GG-6 GitLab Timetracking Research	 Dominik Wyss	DONE 
 GG-5 GitLab Pipeline Setup	 Dominik Wyss	DONE 
 GG-2 GitHub Setup	 Yoris Kucera	DONE 
 GG-87 Document Meetings Structure	 Cedric Cathomas	DONE 
 GG-84 Typst documentation changes	 Cedric Cathomas	DONE 
 GG-86 Document Collaboration as a team	 Cedric Cathomas	DONE 
 GG-7 Typst Documentation Setup	 Cedric Cathomas	DONE 
 GG-8 People, Cost, Time Documentation	 Cedric Cathomas	DONE 
 GG-85 Roles Documentation	 Cedric Cathomas	DONE 
 GG-12 Longtermplan & Milestones	 Nathanael Fässler	DONE 
 GG-9 Jira Setup	 Nathanael Fässler	DONE 
 GG-112 Unity Folder Structure	 Nathanael Fässler	DONE 
 GG-11 Steamworks Registration	 Nathanael Fässler	DONE 
 GG-13 Shorttermplan for first sprint	 Nathanael Fässler	DONE 

Image 4: Sprint 1 Backlog

7.4 Risk Management

This chapter describes potential risks to the project and how we plan to mitigate them. We identified 8 risks, which are listed in the table below. Additionally, these risks are visualized in two risk matrices, one showing the initial risk assessment and the other showing the risk assessment after mitigation strategies are applied.

No.	Risk	Description	Mitigation Strategy
R01	Lack of familiarity with Unity	None of our team members have any significant experience with the Unity Engine. As such, this could heavily impact our development times.	We made ourselves familiar with the engine before the actual start of the project, to get an idea of the workflows.
R02	Unity CI/CD Complexity	We are depending on Unity's build system being robust enough to work in the context of a CI/CD pipeline. We also potentially rely on pre-existing GitHub Actions for Unity, which may not be well-maintained or compatible with our project.	We will set up the CI/CD pipeline very early into the project and did earlier tests. We tried out pre-existing GitHub Actions to make sure they satisfy our needs.
R03	Scope/Feature Creep	As we are developing a game, we run the risk of encountering scope creep, i.e. continuously planning new features, which would result in us eventually not being able to keep up with the workload.	We created a detailed project plan with long-term and short-term planning, along with well defined milestones and a clear MVP.
R04	Bad Code Quality	In large codebases it is easy to end up with badly designed systems that don't fit together with the rest of the code. This would hinder further development.	We do mandatory code reviews before changes can be merged. Additionally, we will set up CI/CD pipelines to automatically run code quality and formatting checks on pull requests.
R05	Team Member is MIA	A team member might be MIA for an extended period of time due to various reasons, such as an illness or damaged hardware. This would reduce our capacity and slow down development.	We can't reduce the possibility of this occurring, but we can distribute work across the remaining team members. We also avoid knowledge monopolies and always aim to share discoveries with the other team members.
R06	Personal Conflicts	There is always the potential for personal conflicts within the team, which would complicate development.	We openly communicate any interpersonal issues and try to resolve conflicts as early as possible.
R07	Data Loss	There is the potential of local storage media failing, resulting in significant data loss.	We mitigate this by frequently committing our changes to GitHub, and keeping untracked files in cloud based locations.
R08	Workload Underestimation	We might be underestimating the amount of work required for specific features and tasks.	We plan with a buffer and try to be conservative with our estimates. We also try to break down large tasks into smaller ones, which are easier to estimate.

Table 21: Risk Analysis

7.4.1 Risk Matrix

The below table shows the risk assessment of the identified risks, before any mitigation strategies are applied. The risks are categorized based on their probability of occurrence and their potential severity.

Probability	Severity			
	Negligible	Marginal	Critical	Catastrophic
Certain				
Likely			R04	R01
Possible		R02/R06		R03/R08
Unlikely				R05
Rare				R07
None				

Table 22: Risk Matrix

7.4.2 Risk Matrix (Mitigated)

The below table shows the risk assessment of the identified risks, after mitigation strategies have been applied.

Probability	Severity			
	Negligible	Marginal	Critical	Catastrophic
Certain				
Likely				
Possible				
Unlikely		R06	R01/R04/R05	R08
Rare	R07	R02		
None				R03

Table 23: Risk Matrix (Mitigated)

8 Time Tracking Report

8.1 Time Tracking

Every member tracks time on Jira on the corresponding task/story. Time spent in meetings is documented in the meeting protocols. The time is logged in 15 minute blocks in the format e.g. 3h 15m.

8.2 Time Statistics

The time is grouped by member and by sprint. Per sprint and person the time is rounded to the nearest hour.

Sprint	Cedric Cathomas	Nathanael Fässler	Yoris Kucera	Dominik Wyss	Total
1	23h	29h	21h	24h	97h
2					
3					
4					
5					
6					
7					
Total	23h	29h	21h	24h	97h

Table 24: Time Statistics Sprint 1

9 Personal Reports

10 Meeting Minutes

10.1 Weekly Scrum / 28.02.2026

Field	Value
Date	28.02.2026
Time	11:00–14:00
Sprint	Sprint 1
Location	Online
Participants	Nathanael Fässler, Dominik Wyss, Yoris Kucera
Excused	Cedric Cathomas
Absent	-
Meeting lead	Yoris Kucera
Minutes by	Nathanael Fässler
Links	-

10.1.1 Agenda

Agenda item	Owner	Result / Decision
GitHub Setup	Yoris	- We switched from GitLab to GitHub The following rules were discussed: - Pull Request requires one approval that is not the same person as the author. - Merging Methods are Merge and Squash. - Every Jira Task corresponds to one branch.
CICD Pipeline Setup	Dominik	- Web build and .exe build succeeded. - Pages Architecture with main and dev branches decided
Timetracking	Dominik	- We won't use GitLab timetracking because we switched to GitHub - Timetracking is in Jira: Everyone tracks time for individual Tasks/Stories.
Unity/Coding Setup	Yoris	Coding Guidelines decided
Steam Release	Nathanael	- Registered on Steamworks - Up to 30 days waiting time until application is processed - Maybe we need to make a contract with OST.
Testing	Nathanael	Unit-Test and Integration-Test Strategy discussed.
Long/Shorttermplan	Nathanael	- Hierarchy: There are Epics which contain Stories. - Everything that is not a game-feature is a task. - Longtermplan with epics according to the Jira Timeline - Shorttermplan per sprint on the Jira board

		• Priority is defined by the order in the backlog in Jira • MVP is defined • We will estimate Tasks/Stories with Story-points • On Jira we can create Bug-Tickets which we will not estimate with Storypoints
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10.1.2 Todos

Todo	Owner	Due
Finish CICD Pipeline	Dominik	06.03.2026
Ask Advisor about contract	Nathanael	04.03.2026
Ask Advisor about project plan	Nathanael	04.03.2026

10.2 Weekly Scrum and Sprint Retrospective / 06.03.2026

Field	Value
Date	06.03.2026
Time	14:00-16:00
Sprint	Sprint 1
Location	Online
Participants	Nathanael Fässler, Dominik Wyss, Yoris Kucera, Cedric Cathomas
Excused	-
Absent	-
Meeting lead	Nathanael Fässler
Minutes by	Cedric Cathomas
Links	-

10.2.1 Agenda

Agenda item	Owner	Result / Decision
Board Review	All	- Walked through all current board tasks - Updated status on completed tasks
Done Tasks and New Functionalities	All	- Reviewed completed work from Sprint 1 - Discussed new functionalities and rules
Open Questions	All	Discussed open questions regarding project setup and workflow
Unity Graphics in Repository	Nathanael Fässler	Only .fbx files are to be committed to the repository for 3D graphics in Unity
Documentation Versioning	Cedric Cathomas	- Tags for documentation versions start with "d" (e.g. d1.1) - Version is updated automatically on pull request to main via GitHub Action
How do we manage smaller tasks that are not worth creating a new task on jira. (small documentation changes)	Dominik Wyss	We create ticket for small backlog tasks that do not need an own Jira item so we can reference it on our pull request and use the name of that ticket for the branch
Decide if we need a separate pdf for protocols	Cedric Cathomas	We decided that there's no use in having a separate page just for meeting minutes, so we have one thing less to worry about

10.2.2 Sprint 1 Retrospective

10.2.2.1 What Went Well

- Good and fast communication
- Good planning on Jira
- Good preplanning visible
- Product Owner did a good job
- Good and understandable selection of planning and working tools
- Pipeline and GitHub are working
- Typst is set up
- General setup went very well

10.2.2.2 What Didn't Go Well

- Problems with GitLab
- Didn't have enough time for some tasks

10.2.2.3 Improvements

- Better time management
- Better definitions of done
- Use GitHub from the beginning

10.2.3 Todos

Todo	Owner	Due
Create ticket for small backlog tasks that do not need an own Jira item	Nathanael Fässler	07.03.2026
Create all table references so documentation looks nice and page breaks are clean	Cedric Cathomas	07.03.2026
Adapt milestone dates to single dates (e.g. 16.02.2026 instead of a time frame)	Cedric Cathomas	07.03.2026
clean up documentation (spaces that are too big)	Cedric Cathomas	07.03.2026
Change date format in documentation to use dots	Cedric Cathomas	07.03.2026
Create task for combined build and documentation pipeline	Nathanael Fässler	07.03.2026
Use figures for images so they appear in the list of figures	Nathanael Fässler	07.03.2026
remove seperate meeting minutes pdf typst code	Cedric Cathomas	07.03.2026

10.3 Sprint Planning / 06.03.2026

Field	Value
Date	06.03.2026
Time	16:00–18:00
Sprint	Sprint 2
Location	Online
Participants	Nathanael Fässler, Dominik Wyss, Yoris Kucera, Cedric Cathomas
Excused	-
Absent	-
Meeting lead	Yoris Kucera
Minutes by	Nathanael Fässler
Links	-

10.3.1 Sprint goal

- Build first working prototype.
- Define Functional Requirements and Non-Functional Requirements.
- Create Domain Model

10.4 Sprint Review (Advisor) / 10.03.2026

Field	Value
Date	10.03.2026
Time	15:00–16:00
Sprint	Sprint 2
Location	Building 8
Participants	Thomas Bocek, Dominik Wyss, Yoris Kucera, Nathanael Fässler, Cedric Cathomas
Excused	-
Absent	-
Meeting lead	Yoris Kucera
Minutes by	Nathanael Fässler
Links	-

10.4.1 Agenda

Agenda item	Owner	Result / Decision
Status of Work	Yoris	Feedback was good
Project Plan	Yoris	Good impression. Advisor liked that we asked him at 02.03.2026 on our initiative
Showcase Pipeline	Dominik	Positive that Dominik sent the pipeline beforehand
Feedback	Thomas	Overall very good
Questions about Grading & Review Schedule	Dominik & Nathanael	- To get a better grade we have to do more than what is required on the checklist. - The group can choose when the Review Meetings take place. - Meeting 5 has to be before 05.06.2026 (final submission)

10.4.2 Todos

Todo	Owner	Due
Update README on GitLab to indicate that project has moved to GitHub	Yoris	20.03.2026
Configure Pipeline to build documentation	Not assigned	20.03.2026
Add AGENTS.md	Cedric Cathomas	20.03.2026

10.4.3 Next dates

Next date	When
Review 2 - Requirements	24.03.2026 15:00

10.5 Weekly Scrum / 13.03.2026

Field	Value
Date	13.03.2026
Time	14:30-15:30
Sprint	Sprint 2
Location	Online
Participants	Cedric Cathomas, Nathanael Fässler, Dominik Wyss, Yoris Kucera
Excused	-
Absent	-
Meeting lead	Dominik Wyss
Minutes by	Yoris Kucera
Links	-

10.5.1 Agenda

Agenda item	Owner	Result / Decision
Time Tracking	Dominik Wyss	- New time tracking report system integrated in Jira (Worklogs Report) - Task does not need to be assigned for time to be tracked
Documentation Build Pipeline	Yoris Kucera	- Pipeline for building documentation on GitHub Actions is set up and working - Automatically deployed to pages on push to dev and main branches.
Domain Model	Yoris Kucera	- First version of domain model created - State machine for enemy created
Functional Requirements	Nathanael Fässler	- Documented Functional Requirements, Actors and Personas - Documented Use Cases - Use Cases formatting could be improved
Player and Camera	Nathanael Fässler	- Player controller implemented and working, controls with WASD - Camera implemented and working, controls with mouse movement - Camera follows player and rotates around it, fixed height and distance
Non-Functional Requirements	Cedric Cathomas	- Documented Non-Functional Requirements - Priority Matrix for NFRs - NFRs grouped by category and numbered (e.g. NFR101, NFR102, NFR403, etc.)
GitHub CI/CD Failing	Dominik Wyss/Yoris Kucera	- Pipeline fails because of lack of disk space on the runner - Possible solutions: ▸ Clean up even more storage ▸ Make repository public ▸ Use self-hosted runner with more storage ▸ Rent server at OST

UI Sketches	Dominik Wyss/ Cedric Cathomas	• Sketches for Main Menu ▸ Scene from the game shown in the background ▸ Buttons for New Game, Continue, Save Files, Settings, Exit • Sketch for New Game menu ▸ Give name to world • Sketch for Save File Menu ▸ List of save files with name and date ▸ Button to delete save file ▸ Show glowcore level? ▸ Separate list for auto saves and manual saves? • Sketch for Settings Menu ▸ Sliders for music and sound effects volume ▸ Slider for mouse sensitivity? ▸ Graphics options? • Sketch for Pause Menu
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10.5.2 Todos

Todo	Owner	Due
Fix WebGL Build Pipeline	Yoris Kucera	20.03.2026
Unify NFRs, FRs and Domain Model	Yoris Kucera	20.03.2026
Adjust README on GitLab	Dominik Wyss	20.03.2026
Agents.md	Cedric Cathomas	20.03.2026
Explicitly address review feedback in Teams	Dominik Wyss	20.03.2026

10.6 Weekly Scrum and Sprint Retrospective / DD.MM.YYYY

Field	Value
Date	DD.MM.YYYY
Time	15:00–16:00
Sprint	Sprint 3
Location	Room / Online
Participants	Team A, Team B
Excused	-
Absent	-
Meeting lead	Scrum Master
Minutes by	Name
Links	Retro board link

10.6.1 What went well

- ...

10.6.2 What didn't

- ...

10.6.3 Todos

Todo	Owner	Due
Introduce PR review rule: 1 reviewer min	Name	DD.MM.YYYY
Add DoD checklist to every ticket	Name	DD.MM.YYYY
Open	Name	DD.MM.YYYY

10.6.4 Retro decisions

- Process decision: ...

10.7 Sprint Planning / DD.MM.YYYY

Field	Value
Date	DD.MM.YYYY
Time	16:00–17:00
Sprint	Sprint 4
Location	Room / Online
Participants	Team A, Team B
Excused	-
Absent	-
Meeting lead	PO / SM
Minutes by	Name
Links	Backlog link

10.7.1 Sprint goal

- ...

10.7.2 Selected backlog (commitment)

ID	Item	Est.	Owner
T-123	Implement login flow	5	Name
T-124	Add CI pipeline	3	Name

10.7.3 Risks / Dependencies

- Dependency: ...
- Risk: ...

10.7.4 Todos

Todo	Owner	Due
Prepare demo script	Name	YYYY-MM-DD
Open		

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